

Finance Club, IIT Roorkee: Open Projects 2026

Project: Implied Volatility Surface Prediction

Predicting missing implied volatility values across strikes and timestamps for options market data



1 Core Problem Statement

Implied volatility is the pricing language of the options market. Incorrect fair-value estimates for implied volatility can materially affect option pricing, risk management, hedging decisions, and trading strategies. The challenge is to build a model that can accurately predict missing values in an implied volatility surface using the available option data across strikes and timestamps.

2 Project Overview

This project focuses on predicting missing implied volatility values in option market data. Participants are provided with a CSV file containing implied volatility observations for options with different strikes at different timestamps. Some of the implied volatility values are missing, and the task is to estimate these values as accurately as possible.

The problem is directly relevant to quantitative finance because implied volatility surfaces are central to options pricing, volatility trading, portfolio risk measurement, and derivatives market-making. A robust model should learn the structure of the volatility surface across both the strike dimension and the time dimension, while handling missing data in a statistically sound and financially meaningful way.

3 Background and Introduction

In options markets, implied volatility represents the market's consensus estimate of future uncertainty embedded in option prices. Rather than quoting option prices directly, traders often compare and communicate value through implied volatility. This makes implied volatility a compact and widely used pricing language for derivatives.

An implied volatility surface describes how implied volatility varies across strike prices and expiries or timestamps. In practice, the surface is rarely perfectly complete. Missing values can arise from illiquid strikes, sparse trading, data-quality issues, bid-ask filtering, or market microstructure noise. Accurately recovering these missing values is important because even small errors in implied volatility can translate into meaningful pricing errors, especially for portfolios with large option exposure.

The project therefore asks participants to combine data science, statistical modelling, and financial intuition. The strongest approaches are expected to respect smoothness and structure in the implied volatility surface while still adapting to local patterns in the data.

4 Objective

Participants must develop a predictive model that fills in missing implied volatility values in the provided option dataset. The objective is to generate predictions that minimize the mean squared error between the submitted values and the hidden ground-truth implied volatility values.

More specifically, participants should:

- Understand the structure of the given option implied volatility dataset.
- Identify and model relationships across strikes, timestamps, and observed implied volatility values.
- Predict the missing implied volatility entries in the required submission format.
- Produce a reproducible notebook that generates exactly the same CSV as the submitted file.

5 Dataset Description

Participants will be provided with a CSV file containing implied volatility values from the Nifty50 options chain across different strikes and timestamps. The dataset includes observed implied volatility values as well as missing entries that must be predicted.

While the exact modelling choice is open-ended, participants may consider the following aspects while exploring the data:

- The cross-sectional relationship between strike and implied volatility at a given timestamp.
- The time-series relationship of implied volatility for the same or nearby strikes.
- Smoothness, curvature, skew, and smile-like behaviour in the implied volatility surface.
- Outliers, stale values, noisy observations, and patterns in missingness.
- Validation strategies that avoid leakage from the hidden target values.
- Prevention of lookahead bias when using timestamp-based features.

6 Suggested Modelling Directions

Participants are free to use any suitable modelling approach, provided the final notebook is reproducible and generates the submitted CSV exactly. Possible approaches include:

- Interpolation and smoothing methods across strikes and timestamps.
- Time-series and panel-data models that exploit nearby observations.
- Feature engineering based on moneyness, strike distance, timestamp ordering, or neighbouring implied volatility values.
- Machine learning models such as gradient boosting, random forests, neural networks, or regularized regression.
- Ensemble methods that combine financial interpolation with data-driven prediction.

The best solutions will likely combine predictive accuracy with an understanding of the implied volatility surface. A purely mechanical model may perform poorly if it ignores the financial structure of option data, while a purely theoretical model may miss dataset-specific patterns.

Participants must ensure that their modelling approach does not introduce lookahead bias. In particular, features should not use information from future timestamps when predicting a missing value at a given timestamp. However, participants are allowed to use available data from other strikes at the same timestamp, since the implied volatility surface is naturally cross-sectional at each point in time.

7 Expected Deliverables

The project is hosted on Kaggle, and all prediction submissions must be made through the Kaggle competition page:

[Kaggle Competition Link](#)

Participants are expected to submit:

- A CSV file containing the predicted implied volatility values in the required Kaggle submission format.
- A notebook that can exactly reproduce the submitted CSV file.
- Clear preprocessing, feature engineering, modelling, and prediction steps inside the notebook.
- Any assumptions, validation approach, or modelling choices needed to make the work understandable and reproducible.

8 Kaggle Leaderboard and Final Evaluation

The judging metric for the challenge is mean squared error (MSE). The goal is to minimize the MSE between the predicted implied volatility values and the hidden true values.

The leaderboard structure is as follows:

- **Public leaderboard:** Shows performance on 30% of the evaluation data.
- **Private leaderboard:** Determines the final ranking using the remaining 70% of the evaluation data.
- **Final reveal:** The private leaderboard will be revealed only after code verification of the top submissions.

Participants should therefore treat the public leaderboard as only an indicative signal. A model that performs well on the public leaderboard may not necessarily perform equally well on the private leaderboard, so robust validation and generalization should be prioritized.

9 Code Collection and Verification

All submitted notebooks must be original, reproducible, and aligned with the CSV submitted on Kaggle. Any plagiarism in code, if found during verification, will lead to disqualification.

A Google Form will be floated later for collecting participants' code files or notebooks and Kaggle IDs. Participants should ensure that the Kaggle ID submitted through the form matches the Kaggle account used for the competition submission.

Important resume verification rule: For resume purposes, the Finance Club will verify the **top 25% of participants, or the top 50 participants, whichever number is higher.**

10 Evaluation Criteria

Submissions will be assessed primarily on predictive performance. However, participants should also ensure that their notebook is clean, reproducible, and aligned with the submitted CSV. A strong submission should demonstrate:

1. **Prediction accuracy:** The model should minimize MSE on the hidden evaluation set.

2. **Reproducibility:** The submitted notebook should exactly reproduce the submitted CSV.
3. **Sound validation:** The modelling workflow should use a reasonable validation strategy and prevent lookahead bias.
4. **Financial intuition:** The approach should reflect the structure of implied volatility surfaces rather than treating the data as unrelated cells.
5. **Code originality:** The code should be original, properly written by the participant or team, and free from plagiarism.
6. **Code quality:** The notebook should be readable, organized, and easy to rerun.